Chapter 4.8 - Control Hazards & Branch Prediction

Comprehensive Revision Flashcards • 69 Key Concepts & Practice Questions

#1

Q1: What is a control hazard in a processor pipeline?

Answer: It is a delay in determining the proper instruction to fetch, caused by a conditional branch instruction.

#2

Q2: In the non-optimized five-stage pipeline described, in which stage is the decision about whether to branch made?

Answer: The decision is made in the MEM (Memory Access) pipeline stage.

#3

Q3: Without any optimization, how many sequential instructions following a branch are fetched before the branch decision is made in the MEM stage?

Answer: Three sequential instructions are fetched and begin execution.

#4

Q4: The technique of discarding instructions in a pipeline, usually due to an unexpected event like a mispredicted branch, is called _____.

Answer: flushing

#5

Q5: What is the simplest static branch prediction scheme mentioned in the text?

Answer: The 'Assume Branch Not Taken' scheme, where execution continues down the sequential instruction stream.

#6

Q6: In the 'Assume Branch Not Taken' scheme, what happens if the conditional branch is actually taken?

Answer: The instructions that were being fetched and decoded must be discarded, and execution continues at the branch target.

#7

Q7: How are instructions discarded or flushed when a branch is mispredicted in the 'Assume Not Taken' scheme?

Answer: The original control values for the instructions in the IF, ID, and EX stages are changed to 0s.

#8

Q8: What is the primary method discussed to improve conditional branch performance by reducing the cost of a taken branch?

Answer: Moving the conditional branch execution earlier in the pipeline, from the MEM stage to the ID stage.

#9	**Q9: What are the two actions that must occur earlier to move conditional branch execution to the ID stage?**
	Answer: Computing the branch target address and evaluating the branch decision must both occur earlier.

#10	**Q10: What hardware modification is needed in the ID stage to calculate the branch target address earlier?**
	Answer: The branch adder is moved from the EX stage to the ID stage.

#11	**Q11: What hardware modification is needed in the ID stage to evaluate the branch decision for 'branch if equal' earlier?**
	Answer: An equality test unit is added to compare two register values during the ID stage.

#12	**Q12: When branch execution is moved to the ID stage, what new hardware complication arises regarding data dependencies?**
	Answer: New forwarding logic is required, as the equality test unit in ID may need results from later pipeline stages (EX/MEM or MEM/WB).

#13	**Q13: If an ALU instruction immediately precedes a branch that depends on its result, what is required when branch execution is in the ID stage?**
	Answer: A stall will be required because the ALU result is produced in the EX stage, after the branch's ID cycle.

#14	**Q14: If a load instruction is immediately followed by a conditional branch that depends on its result, how many stall cycles are needed if branch execution is in the ID stage?**
	Answer: Two stall cycles are needed, as the load result is not available until the end of the MEM cycle.

#15	**Q15: By moving branch execution to the ID stage, the penalty of a taken branch is reduced to how many instructions?**
	Answer: The penalty is reduced to only one instruction, which is the one currently being fetched.

#16	**Q16: What is the purpose of the `IF.Flush` control line?**
	Answer: It zeros the instruction field of the IF/ID pipeline register, effectively transforming the fetched instruction into a nop.

#17	**Q17: Prediction of branches at runtime using runtime information is known as _____.**
	Answer: dynamic branch prediction

#18	**Q18: What is a branch prediction buffer, also known as a branch history table?**
	Answer: It is a small memory indexed by the lower portion of a branch instruction's address, containing bits that indicate if the branch was recently taken.

#19	**Q19: What is the primary performance shortcoming of a simple 1-bit dynamic branch prediction scheme?**
	Answer: If a branch is almost always taken (like in a loop), it will be mispredicted twice for each full loop execution rather than just once.

#20	**Q20: For a loop branch that is taken nine times and not taken once, what is the prediction accuracy using a 1-bit scheme?**
	Answer: The accuracy is 80%, with two incorrect predictions (on the first and last iterations) and eight correct ones.

#21	**Q21: Why does a 1-bit predictor mispredict the first iteration of a loop that was just completed?**
	Answer: The prediction bit was flipped to 'not taken' on the final, exiting iteration of the previous loop execution.

#22	**Q22: How does a 2-bit branch prediction scheme improve over a 1-bit scheme?**
	Answer: A prediction must be wrong twice before the prediction state is changed, leading to only one misprediction for highly regular loops.

#23	**Q23: In a 2-bit counter-based predictor, what action is taken when the prediction is accurate?**
	Answer: The counter is incremented (or moved towards a more strongly-taken/not-taken state).

#24	**Q24: In a 2-bit counter-based predictor, what action is taken when the prediction is inaccurate?**
	Answer: The counter is decremented (or moved towards the weakly-taken/not-taken state, and eventually flips).

#25	**Q25: What structure caches the destination PC or destination instruction for a branch to reduce the taken-branch penalty?**
	Answer: A branch target buffer.

#26	**Q26: How does a branch target buffer differ from a simpler branch prediction buffer?**
	Answer: It is usually organized as a cache with tags, making it more costly, and it stores the target address or instruction, not just a prediction bit.

#27	**Q27: What is a correlating predictor?**
	Answer: A branch predictor that combines the local behavior of a branch with global information about recently executed branches.

#28	**Q28: What is a tournament branch predictor?**
	Answer: A predictor that uses multiple prediction schemes and a selection mechanism to choose the best one for a given branch.

#29	**Q29: What type of instruction can be added to an instruction set to reduce the number of conditional branches?**
	Answer: Conditional move instructions.

#30	**Q30: What does the ARMv8 `CSEL` instruction do?**
	Answer: It conditionally selects one of two source registers to copy to a destination register based on a condition code.

#31	**Q31: For a branch with 5% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The predict-not-taken scheme is best, as it will be correct 95% of the time.

#32	**Q32: For a branch with 95% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The predict-taken scheme is best, as it will be correct 95% of the time.

#33	**Q33: For a branch with 70% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The dynamic predictor is best, as its 90% accuracy is higher than predict-taken (70%) or predict-not-taken (30%).

#34	**Q34: What is the name for a pipeline hazard involving conditional branches?**
	Answer: It is called a control hazard or a branch hazard.

#35	**Q35: In the unoptimized five-stage pipeline described, during which stage is the decision about whether to branch made?**
	Answer: The branch decision occurs during the MEM (memory access) pipeline stage.

#36	**Q36: The delay in determining the proper instruction to fetch due to a branch instruction is called a _____ or branch hazard.**
	Answer: control hazard

#37	**Q37: Why is the section on control hazards presented as shorter than the one on data hazards?**
	Answer: Control hazards are simpler to understand, occur less frequently, and have less effective resolution schemes compared to forwarding for data hazards.

#38	**Q38: What is a simple static branch prediction scheme that improves upon stalling?**
	Answer: The 'assume branch not taken' scheme, where execution continues down the sequential instruction stream.

#39	**Q39: What action is taken to discard instructions in a pipeline when a branch is mispredicted?**
	Answer: The original control values for the incorrect instructions are changed to 0s.

#40	**Q40: Term: Flush**
	Answer: Definition: To discard instructions in a pipeline, usually due to an unexpected event like a mispredicted branch.

#41	**Q41: In the unoptimized pipeline, which stages must be flushed when a branch taken in the MEM stage is discovered?**
	Answer: The instructions in the IF, ID, and EX stages must be flushed.

#42	**Q42: What is one way to improve conditional branch performance by reducing the cost of a taken branch?**
	Answer: Move the conditional branch execution earlier in the pipeline, such as to the ID stage.

#43	**Q43: What are the two actions that must occur earlier to move branch execution from the MEM stage to the ID stage?**
	Answer: Computing the branch target address and evaluating the branch decision.

#44	**Q44: How is the branch target address calculation moved to the ID stage?**
	Answer: The branch adder is moved from the EX stage to the ID stage, using the PC and immediate field from the IF/ID register.

#45	**Q45: What is the more difficult part of moving branch execution to the ID stage?**
	Answer: Moving the branch decision itself, which involves comparing two register values during the ID stage.

#46	**Q46: Moving the branch test to the ID stage introduces the need for new _____ and _____ hardware.**
	Answer: forwarding, hazard detection

#47	**Q47: If branch comparison is moved to ID, from which pipeline registers can the bypassed source operands come?**
	Answer: The bypassed operands can come from either the EX/MEM or MEM/WB pipeline registers.

#48	**Q48: If branch execution is moved to the ID stage, what happens if an ALU instruction immediately preceding a branch produces an operand for the branch test?**
	Answer: A data hazard occurs, and a one-cycle stall is required.

#49	**Q49: If branch execution is moved to the ID stage, what is the penalty for a load instruction immediately followed by a branch that depends on the load result?**
	Answer: A two-cycle stall is needed.

#50	**Q50: What is the primary benefit of moving conditional branch execution to the ID stage?**
	Answer: It reduces the penalty of a taken branch to only one instruction.

#51	**Q51: What is the purpose of the IF.Flush control line?**
	Answer: It zeros the instruction field of the IF/ID pipeline register to flush the instruction currently in the IF stage.

#52	**Q52: Clearing the IF/ID register with IF.Flush transforms the fetched instruction into a _____, an instruction that has no action.**
	Answer: nop

#53	**Q53: What is the term for predicting branches at runtime using runtime information?**
	Answer: Dynamic branch prediction.

#54	**Q54: Is a prediction from a simple branch prediction buffer guaranteed to be for the correct branch instruction?**
	Answer: No, another branch with the same low-order address bits may have set the prediction bit, but this does not affect correctness.

#55	**Q55: What happens in a 1-bit dynamic prediction scheme when a prediction turns out to be wrong?**
	Answer: The incorrectly predicted instructions are deleted, the prediction bit is inverted, and the proper sequence is fetched.

#56	**Q56: What is the performance shortcoming of a simple 1-bit prediction scheme, especially with loops?**
	Answer: It can mispredict twice for a frequently-taken loop: once on the first iteration and once on the final, non-taken iteration.

#57	**Q57: For a loop branch that is taken 9 times and not taken once, what is the prediction accuracy of a 1-bit scheme?**
	Answer: The accuracy is 80%, with two incorrect predictions (first and last iteration) and eight correct ones.

#58	**Q58: How does a 2-bit branch prediction scheme improve upon a 1-bit scheme?**
	Answer: A prediction must be wrong twice before it is changed, making it more resilient to single changes in branch behavior.

#59	**Q59: In a 2-bit prediction scheme, a branch that strongly favors taken or not taken will be mispredicted how many times when its pattern is briefly interrupted?**
	Answer: It will be mispredicted only once.

#60	**Q60: A 2-bit scheme is a general instance of a _____ predictor, which is incremented on an accurate prediction and decremented otherwise.**
	Answer: counter-based

#61	**Q61: Term: Branch Target Buffer**
	Answer: Definition: A structure that caches the destination PC or destination instruction for a branch to reduce or eliminate the taken-branch penalty.

#62	**Q62: What kind of predictor combines the local behavior of a branch with global information about recently executed branches?**
	Answer: A correlating predictor.

#63	**Q63: Term: Correlating Predictor**
	Answer: Definition: A branch predictor that combines local behavior of a particular branch and global information about the behavior of some recent number of executed branches.

#64	**Q64: Term: Tournament Branch Predictor**
	Answer: Definition: A predictor with multiple prediction mechanisms for each branch and a selector that chooses which one to use based on which has been more accurate.

#65	**Q65: What is an architectural alternative to conditional branches that can reduce their frequency?**
	Answer: Conditional move instructions, which conditionally change a destination register's value instead of the PC.

#66	**Q66: What is the function of the ARMv8 CSEL instruction?**
	Answer: It conditionally selects one of two source registers to copy to a destination register based on the condition codes.

#67	**Q67: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 5% frequency?**
	Answer: Predict not taken is best, as it is correct 95% of the time, yielding a lower average penalty than the 90% accurate dynamic predictor.

#68	**Q68: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 95% frequency?**
	Answer: Predict taken is best, as it is correct 95% of the time, yielding a lower average penalty than the 90% accurate dynamic predictor.

#69	**Q69: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 70% frequency?**
	Answer: Dynamic prediction is best, as its 90% accuracy is better than both predict taken (70%) and predict not taken (30%).